Paper Conclusion

For

(Information Retrieval and Machine Learning Methods for Academic Expert Finding)

Paper Link: Information Retrieval and Machine Learning Methods for Academic Expert Finding

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1. Introduction.

Sometimes we need an expert to help us with a problem or a specific topic, but we can't easily find them. So, in this paper author use concept of Expert Finding, It means finding the right person with expertise in a specific topic based on our request.

A. Background of the research topic.

The Academic expert system has many information about Doctors, Researchers, etc.

Such as: Their academic profile, CV, Articles or publications, Keywords related to their field

and so on, the user enters information about his problem like abstract and system give me the

best experts.

This system work on two datasets and compare between two modern techniques:

Information retrieval and Machine learning.

B. Motivation for selecting the paper.

This paper is selected because include an important problem in Academic Research and

compares different techniques, including IR and ML methods

C. Research problem addressed by the paper.

The paper study academic-expert-finding methods by utilizing information retrieval (IR) and machine learning (ML) With classical and deep learning models.

2. Overview of the Selected Paper.

A. Research objectives and contributions.

Create a smart system that recommends the best expert based on your needs by using

their research and analyze it with IR or ML.

B. Type of research.

This paper is an Experimental Research

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3. Research Methodology Analysis

A. Description of the Methodology

The paper tackles academic expert finding through two fundamentally different lenses, and the decision to run both in parallel rather than picking one is itself a contribution worth noting.

The IR approach builds textual profiles for each researcher from their published papers. Instead of dumping everything into one big profile or treating every paper separately, the authors cluster papers thematically first. Each cluster becomes a subprofile representing a specific research area. A query comes in, gets matched against these subprofiles, and researchers get ranked. The final score per author is computed by fusing their subprofile scores. This middle-ground design is smart — it avoids the known problems of both extremes. A single merged profile buries niche expertise under dominant topics, while one-subprofile-per-paper fragments the signal too aggressively. Clustering splits the difference.

The ML approach is structurally simpler to describe: treat it as text classification. Each author is a class, each paper is a training instance, and the classifier learns to predict authorship from textual content. At inference, a new text expressing some expertise need gets fed in, and the model returns a ranked list of probable authors. The tricky part is co-authorship. Since many papers have multiple authors, the authors handle this by duplicating each training instance once per author, giving each copy a single label. It works, but it inflates the training set and quietly introduces label noise — a limitation the paper touches on without fully confronting.

B. Data Sources, Datasets, Tools, Models, and Algorithms

Datasets

Both datasets come from the biomedical domain, which is both a strength and a ceiling. PMSC-UGR is broad — general biomedicine, 341,520 papers, 9,453 authors, disambiguated via ORCID codes. CORD-19 is narrow — COVID-19 research, 130,477 papers, 7,389 authors, with no disambiguation, so the authors had to clean it manually by filtering out initials-only name entries. Having two datasets with different topical breadths, sizes, and data quality levels is genuinely useful for stress-testing the methods. That said, both are biomedical. Whether any of this holds in law, engineering, or the humanities is completely unknown.

Both datasets contain titles, abstracts, and keywords per article. A minimum publication threshold was enforced — at least 10 papers per author in PMSC-UGR and 20 in CORD-19 — to ensure the models had enough signal to learn from.

Tools

The implementation stack is standard and reproducible. Scikit-learn handled classical ML and K-means. Gensim was used for LDA. Apache Lucene handled indexing and retrieval, using a Language Model with Jelinek–Mercer smoothing. Keras on top of TensorFlow covered the deep learning side.

Models and Algorithms

On the IR side, clustering was done with both K-means and LDA across a wide range of K values: 10, 20, 54, 110, 239, and 390. Two edge cases were also tested — Joint, where all of an author's papers collapse into one profile, and Atomic, where every paper is its own subprofile. Score fusion used either CombMax or CombLgDCS depending on which worked better per dataset. Testing this many K values is one of the stronger design choices in the paper. It turns out the method is largely insensitive to K, which is a practically useful finding — it means you do not need to tune this carefully in deployment.

On the ML side, six classical models were tested: Decision Tree, KNN, Logistic Regression, Naive Bayes, Random Forest, and SGD. Four deep learning models were added: MLP, CNN, LSTM, and BiLSTM. Text was represented either as raw TF-IDF vectors or as SBERT embeddings via the fine-tuned sentence-transformers/allenai-specter model. BERT was tested too but dropped after consistently underperforming SBERT. Using a biomedical-domain SBERT model is the right call given the data, though the paper never asks whether a general-purpose model would close the gap — a small but real omission.

C. Experimental Setup and Evaluation Strategy

Splitting

PMSC-UGR was split temporally: papers from 2007 to 2015 for training, 2016 for testing. This mirrors real deployment conditions — you never recommend experts based on papers that do not exist yet. CORD-19 could not be split this way because its papers are too compressed in time, so an 80/20 stratified random split was used instead.

Preprocessing

Stopwords were removed and stemming applied throughout. For clustering input, terms appearing in fewer than 750 articles or more than 90% of the corpus were discarded, and only the top 5,000 most frequent remaining terms were kept. Importantly, this reduction applied only to the clustering step. The subprofiles themselves retained the full original vocabulary.

Evaluation

Each multi-author test article was duplicated once per author, so every test instance has exactly one **ground-truth** label. The metric used is top_n accuracy at three cutoffs — top_1, top_5, and top_10. Under this setup, top_1 equals standard accuracy, and top_k is equivalent to both recall@k and precision@k.

Three cutoffs make sense at this scale. With up to 9,453 candidate authors, finding the right person even in the top 10 is meaningful. The missing piece is a rank-sensitive metric like MRR or NDCG. Top_n treats a correct answer at position 2 the same as one at position 10 — which is fine for a rough comparison but loses information about how confidently and consistently the models rank the right author near the top.

Hardware and Reporting

Experiments ran on an Intel Xeon Gold 6230 @ 2.10 GHz, 20 cores, 314 GB RAM. Only results without feature selection were reported, since applying it consistently hurt performance. Only SBERT was reported for transformer representations. For deep learning models, only the best configuration per model was included. The last point is the weakest — reporting peak performance without stability analysis makes it hard to know whether those numbers are reproducible or just the best of several noisy runs.

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4. Results and Findings

A. Key results reported in the paper.

After testing both IR and ML methods, I noticed that IR models gave better and more stable results.

K-means worked better than LDA, and changing K didn't affect results much.

Also, using sub profiles helped improve accuracy and made it easier to find the correct expert.

For ML models, the results were not the same. Some models like MLP performed well, but others did not.

In general, IR methods were better than ML, and the results were similar on another dataset.

Table 1. Top_n results obtained for PMSC-UGR with the IR-based models.

K	top_1	top_5	top_10	top_1	top_5	top_10
		Using K-means			Using LDA	
10	0.3566	0.5874	0.6644	0.3558	0.5527	0.6316
20	0.3565	0.5861	0.6647	0.3555	0.5585	0.6358
54	0.3596	0.5863	0.6623	0.3563	0.5617	0.6405
110	0.3587	0.5855	0.6587	0.3564	0.5645	0.6434
390	0.3580	0.5812	0.6566	0.3580	0.5692	0.6454
Atomic	0.3502	0.5706	0.6453			
Joint	0.3518	0.5847	0.6659			

Key Result:

The results were generally consistent even when changing the number of clusters (K).

Evaluation:

Top-1, Top-5, and Top-10 accuracy.

Strengths:

The performance is stable.

Not highly affected by changing K.

Main Strength:

K-means performed better than LDA in most cases.

->The difference between methods was more noticeable in Top-5 and Top-10

->Using subprofiles helped improve the performance and made the system better at finding the correct expert.

Table 2. Top_n results obtained for PMSC-UGR with the ML-based models.

Model	top_1	top_5	top_10	top_1	top_5	top_10
		Using text			Using SBERT	
DT	0.0537	0.0613	0.0622	0.0227	0.0501	0.0703
NB	0.0665	0.1438	0.1910	0.1681	0.3641	0.4615
KNN	0.0683	0.1620	0.1624	0.2428	0.4703	0.4706
RF	-	-	-	0.1181	0.2344	0.3002
SGD	0.2272	0.4188	0.4839	0.1785	0.2648	0.2667
LR	0.1948	0.3696	0.4470	0.2707	0.5010	0.5918
MLP	0.3441	0.5791	0.6621	0.2365	0.4600	0.5520
CNN	0.1425	0.2888	0.3600			
LSTM	0.0821	0.2196	0.3111			
BiLSTM	0.0894	0.2417	0.3331			

Key Result:

ML models showed different performance levels.

Evaluation:

Top-1, Top-5, Top-10.

Strengths:

Some models performed well.

SBERT improved results.

Main Strength:

MLP achieved the best results.

➔ Some ML models performed better than others, and MLP gave the best results among them.

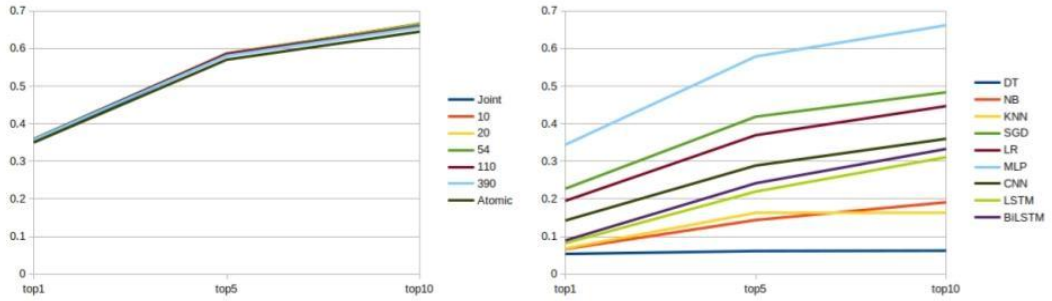

Figure 3. Top_n results for PMSC-UGR for the models based on IR using K-means (**left**) and ML using text (**right**).

Key Result:

IR models outperformed ML models.

Evaluation:

Top-k comparison.

Strengths:

More consistent.

Higher accuracy.

Main Strength:

IR models are more reliable.

➔ IR models performed better than ML models and gave more reliable results.

Table 3. Top_n results obtained for CORD-19 with the IR-based models.

K	top_1	top_5	top_10	top_1	top_5	top_10
		Using K-means			Using LDA	
10	0.1760	0.2812	0.3228	0.1717	0.2791	0.3218
20	0.1792	0.2823	0.3244	0.1779	0.2813	0.3234
29	0.1778	0.2834	0.3240	0.1787	0.2840	0.3248
110	0.1861	0.2893	0.3285	0.1853	0.2922	0.3316
239	0.1901	0.2919	0.3279	0.1883	0.2928	0.3313
Atomic	0.2006	0.3047	0.3466			
Joint	0.1717	0.2874	0.3351			

Table 4. Top_n results obtained for CORD-19 with the ML-based models.

Model	top_1	top_5	top_10	top_1	top_5	top_10
		Using text			Using SBERT	
DT	0.0602	0.0745	0.0758	0.0093	0.0239	0.0359
NB	0.0166	0.0372	0.0504	-	-	-
KNN	0.0389	0.1304	0.1308	0.0835	0.2241	0.2245
RF	-	-	-	0.0363	0.0714	0.0929
SGD	0.1308	0.2397	0.2798	0.0614	0.0967	0.0983
LR	0.0638	0.1275	0.1659	0.1084	0.2162	0.2704
MLP	0.1852	0.3121	0.3696	0.0954	0.1939	0.2401
CNN	0.0486	0.1097	0.1495			
LSTM	0.0362	0.0927	0.1322			
BiLSTM	0.0401	0.1006	0.1411			

Key Result:

Results were similar to the first dataset.

Evaluation:

Top-1, Top-5, Top-10.

Strengths:

Consistent results.

Works on different datasets.

Main Strength:

The model generalizes well.

Final Findings

Key Result:

IR methods gave better results than ML in most cases.

Evaluation:

Based on all experiments.

Strengths:

More stable results

Easier to use

Main Strength:

IR methods worked better overall.

→ I noticed that IR methods were more reliable and gave better results than ML.

B. Performance metrics or evaluation criteria used.

In this paper, they used some formulas to give the results. They focused on the following:

They employed the formula

$$\mathbf{K} = \sqrt{\frac{m}{2}}$$

where m represents the number of articles

in the training set. Consequently, for PMSC-UGR and CORD-19, we obtained K values of 390 and 239, respectively.

Another approach they considered was

$$K = \frac{m * t}{e}$$

where t corresponds to

the number of distinct terms appearing in the articles within the training set, and e

represents the number of non-zero entries in the document-term matrix. For PMSC-UGR and CORD-19,

C. Main strengths of the proposed approach or study.

It didn't depend on just one method, it used both IR and ML, so the comparison was clear and helpful.

Also, they tried the models on more than one dataset, and this made the results feel more reliable, not just based on one case.

The idea of using sub profiles, because it made the representation of each author better and helped in improving the accuracy.

Another good point is that the evaluation method was simple (top-k accuracy), so it was easy to understand the results and compare between models.

D. Limitations and Assumptions

The biggest problem of this research is the biomedical datasets used. Since results are obtained only on them, conclusions are not generalizable to other fields. Furthermore, the authors of a paper are taken to be the correct experts on its topics, and while this satisfies the practical constraints of the experiment, oversimplifies reality, where the authors may be quite representative of the needed expertise, but not the only experts, who would be assigned to a paper. The methodology to evaluate

the method thus further worsens the situation, as it reduces to it a simple author's prediction task, where top-k accuracy is merely measured.

Another important shortcoming is the limited selection of features. The models are limited to analyzing text (titles, keywords, abstracts), since they are not provided with relevant information like citation graphs, co-authorship networks, or the state of research activity over time. Therefore, the experiment does not fully reflect real systems that identify the most knowledgeable experts as these other indicators would influence results. Furthermore, machine learning algorithms were largely used in their default configuration, which can impact performance.

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5. Critical Reflection and Future Directions

A. The critical evaluation.

On a more critical note, the paper is methodologically sound and contributes valuable results, particularly in providing evidence of the robustness and efficiency of the IR-based methods, although the research seems to prefer IR methods over sophisticated machine learning techniques only because they are easier and less costly to implement than deep learning algorithms and transformers. It should be further validated in solving the real-world problem with a focus on conducting a user study or implementing the system into a real-world application.

B. Suggested improvements or future research directions.

There are many ways this experiment can be improved when repeated in the future. Firstly, hybrid approaches of combining textual features with network-based features, such as coauthor ship and citation networks should be used, as results show that they have performance gains over text only approaches. Secondly, machine

learning models should be optimized better, for both existing models and modern transformer models, to see if machine learning approaches, then outperform information retrieval ones. Thirdly, the problem should be correctly recognized as the multi-label problem that it is in reality - research is collaborative! It should also take account of temporal factors that make certain expertise more relevant at that time period, be repeated in other academic domains to generalize the results.

In this study, They recommended experts in a specific topic even if they actively worked in that area 20 years ago, provided that their historical work aligned with the current information need. However, it may be advantageous to prioritize experts whose publications in that same topic are more recent, as this suggests they are more likely to be up-to-date.

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